

# Transformer

- Attention (Self-Attention)
- Transformer Architecture
- Pre-training and Fine-tuning
- Transformer Applications



Prompt

Write an email apologizing to Sarah  
for the tragic gardening mishap.  
Explain how it happened.



Transformer LLM



Generation

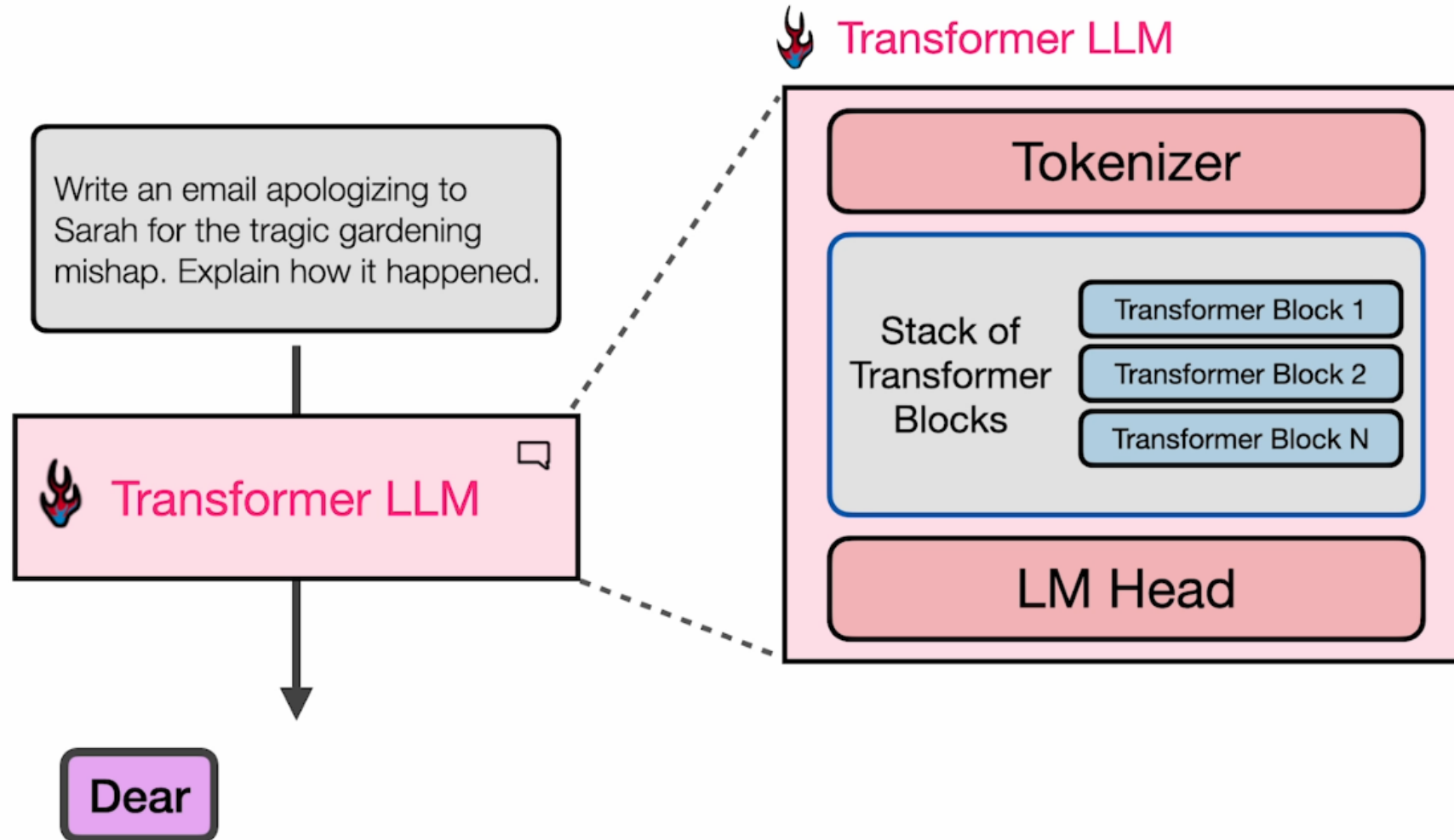
Dear

Sarah

.

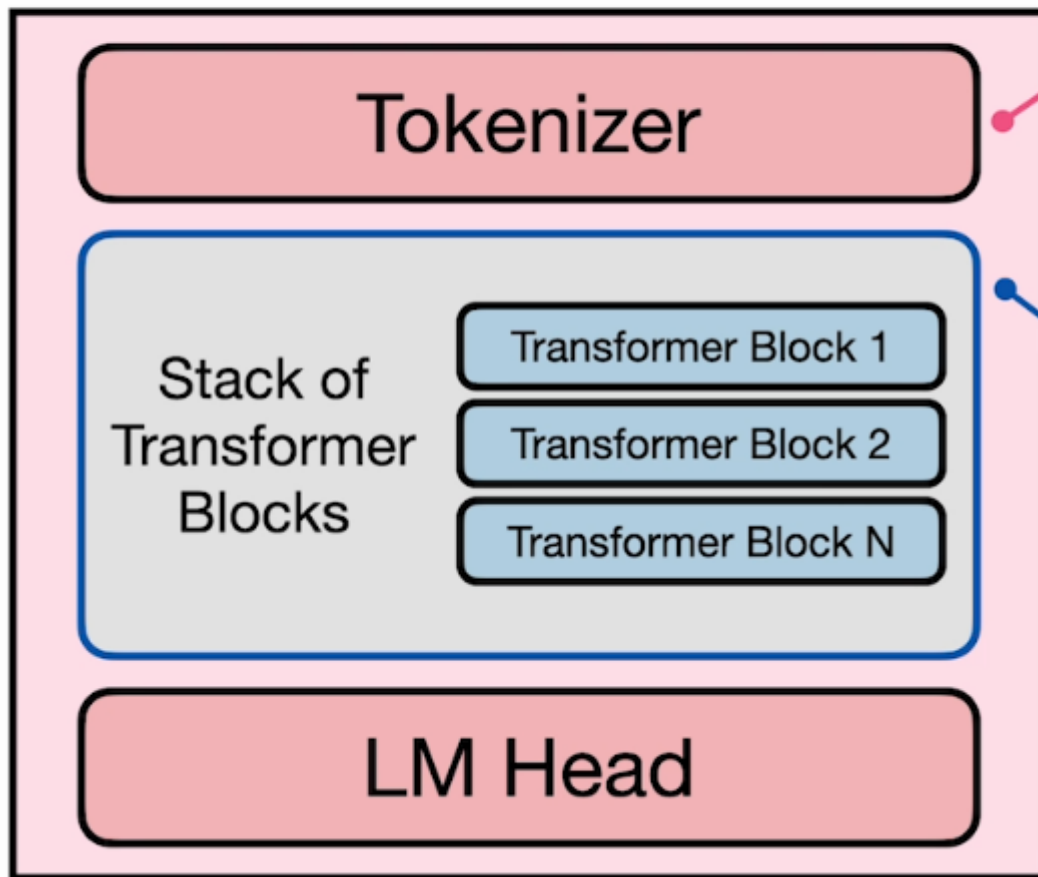
\n

# Three major components: Tokenizer, Transformer Blocks, and LM Head





## Transformer LLM



### Token vocabulary

| Token ID | Token    |
|----------|----------|
| 0        | !        |
| 1        | "        |
| ...      | ...      |
| 50,000   | Zyzzzyva |

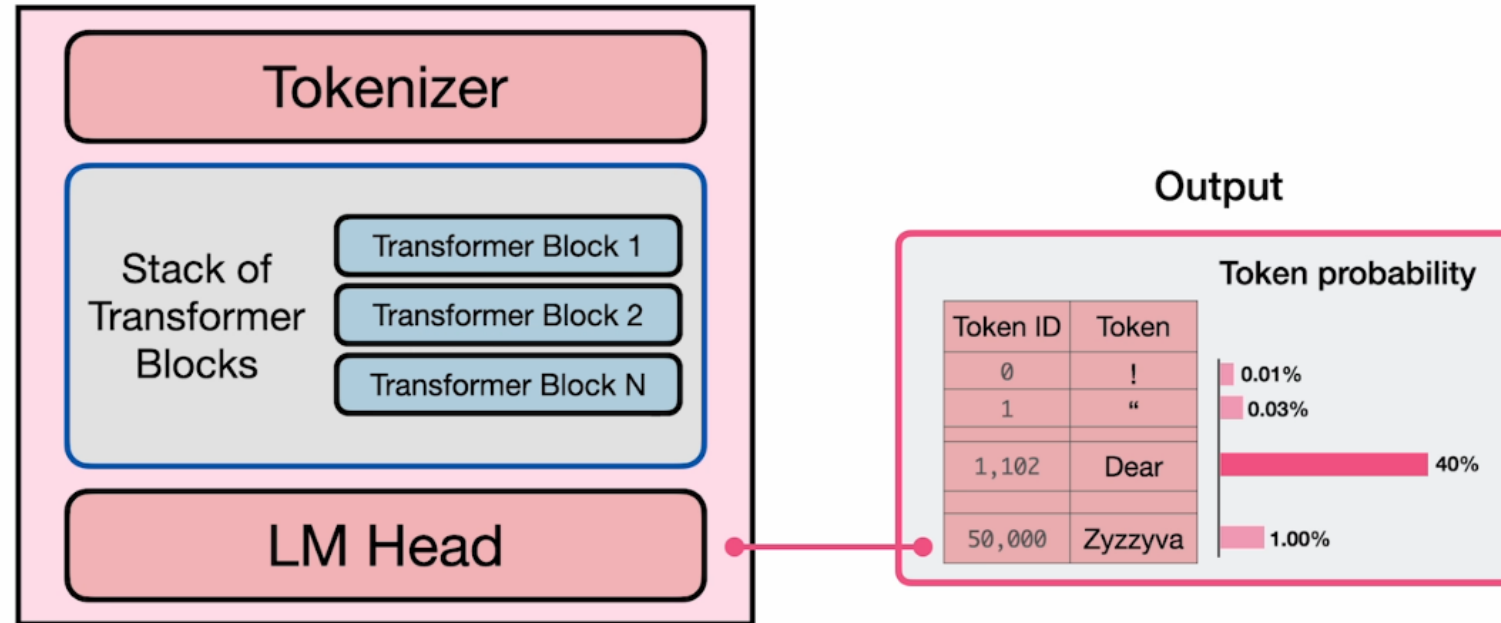
### Token embeddings

|        |                                              |
|--------|----------------------------------------------|
| 0      | <div><div></div><div></div><div></div></div> |
| 1      | <div><div></div><div></div><div></div></div> |
| ...    | ...                                          |
| 50,000 | <div><div></div><div></div><div></div></div> |

The LM head scores the best (most probable) next token to output



## Transformer LLM



Write an email apologizing to Sarah for the tragic gardening mishap. Explain how it happened.



Transformer LLM

### Output Token Probabilities

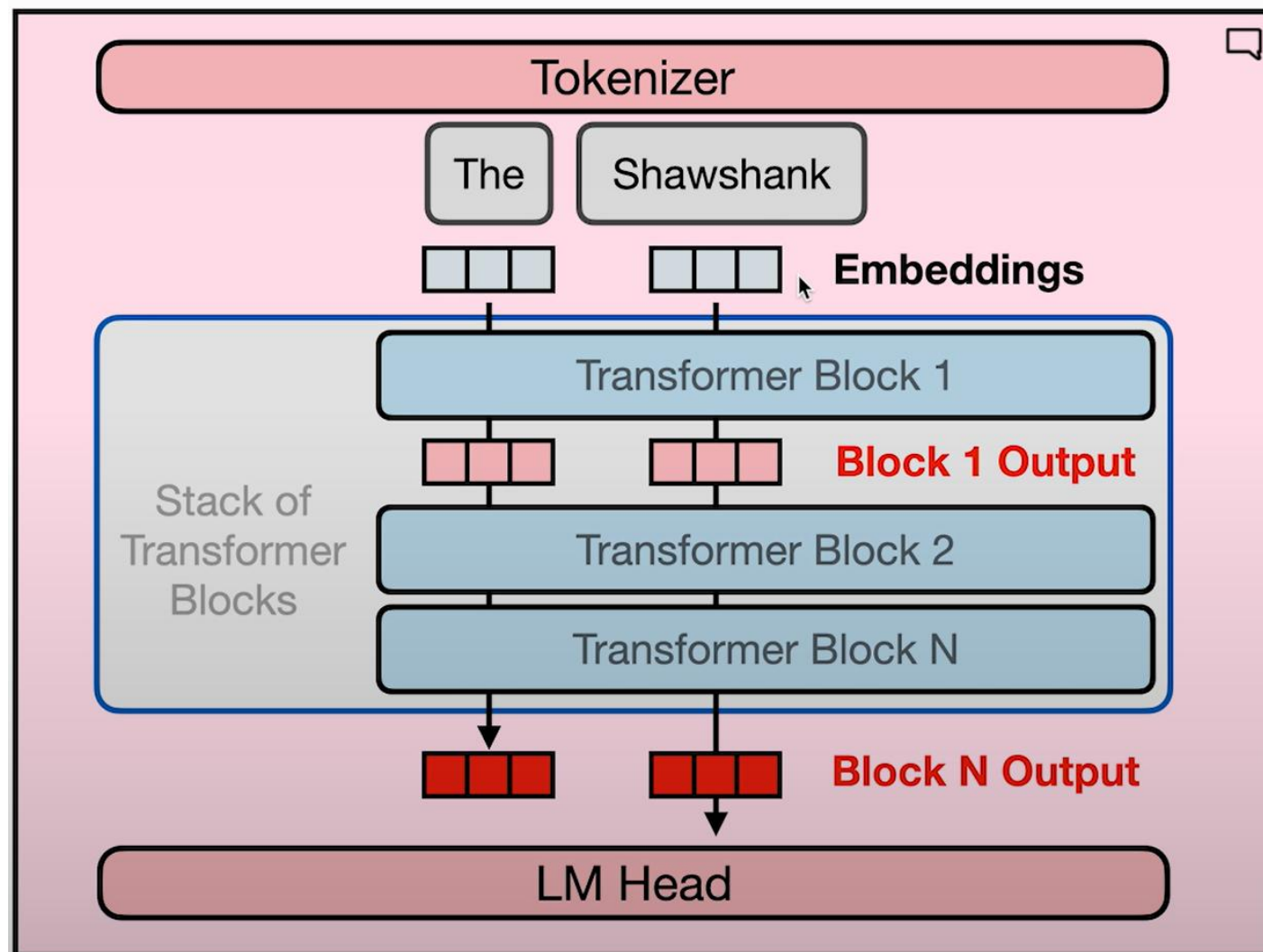
|       |     |
|-------|-----|
| Dear  | 40% |
| Title | 13% |
| To    | 8%  |
| Hi    | 2%  |
| ...   |     |

Decoding strategy

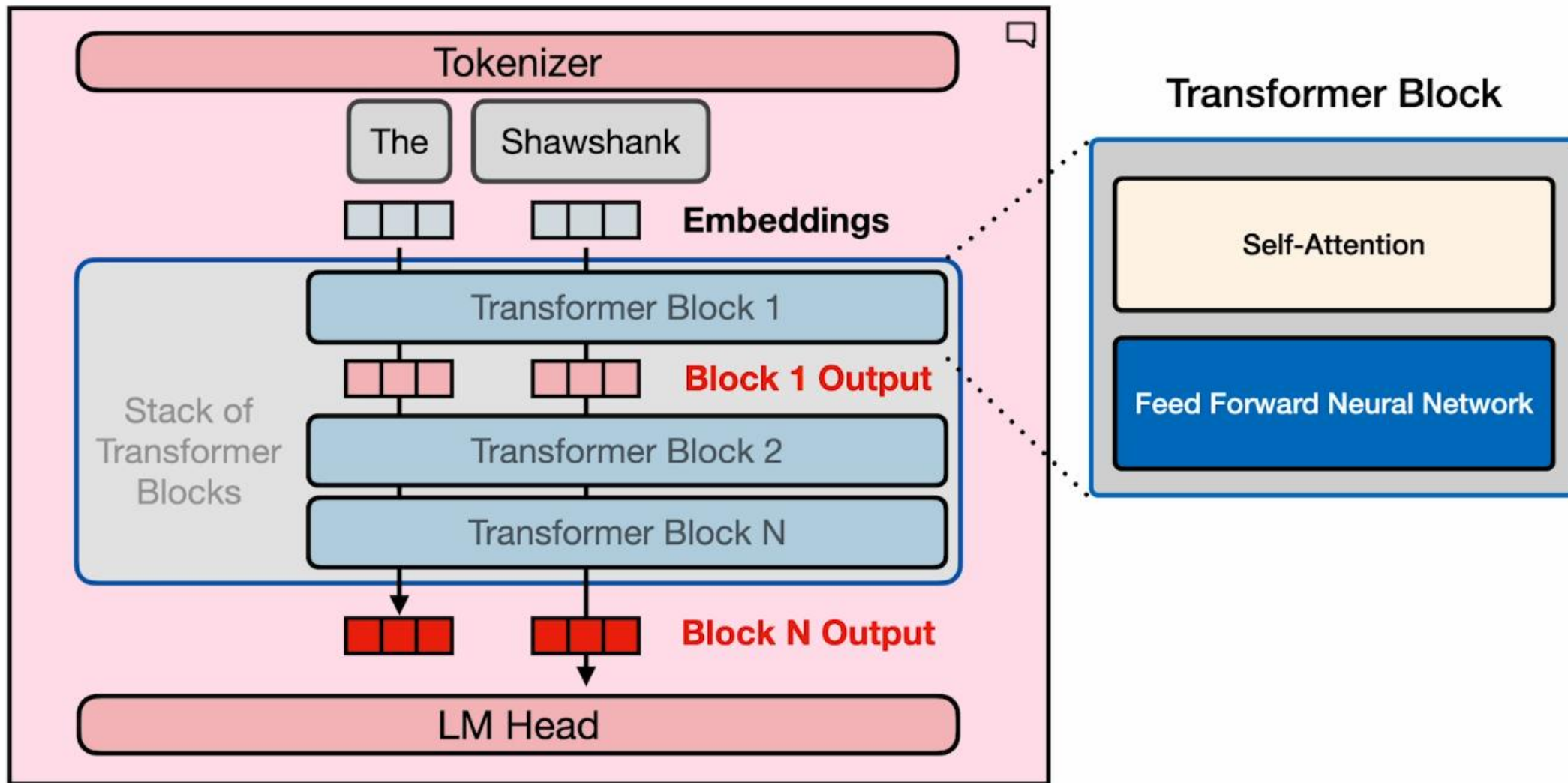
Dear

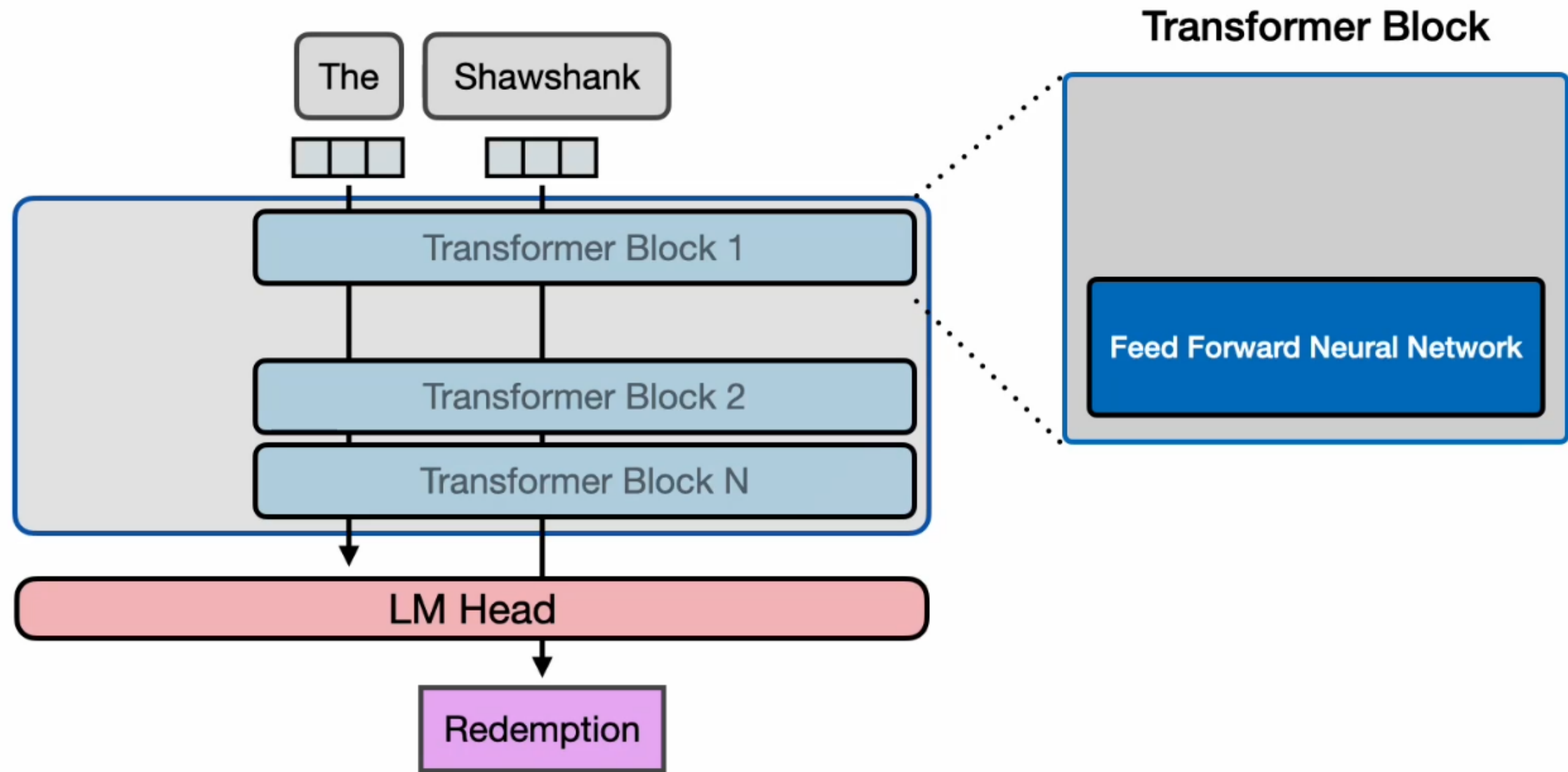
- Choose top\_p
- Add randomness (temperature>0)

## Transformer LLM

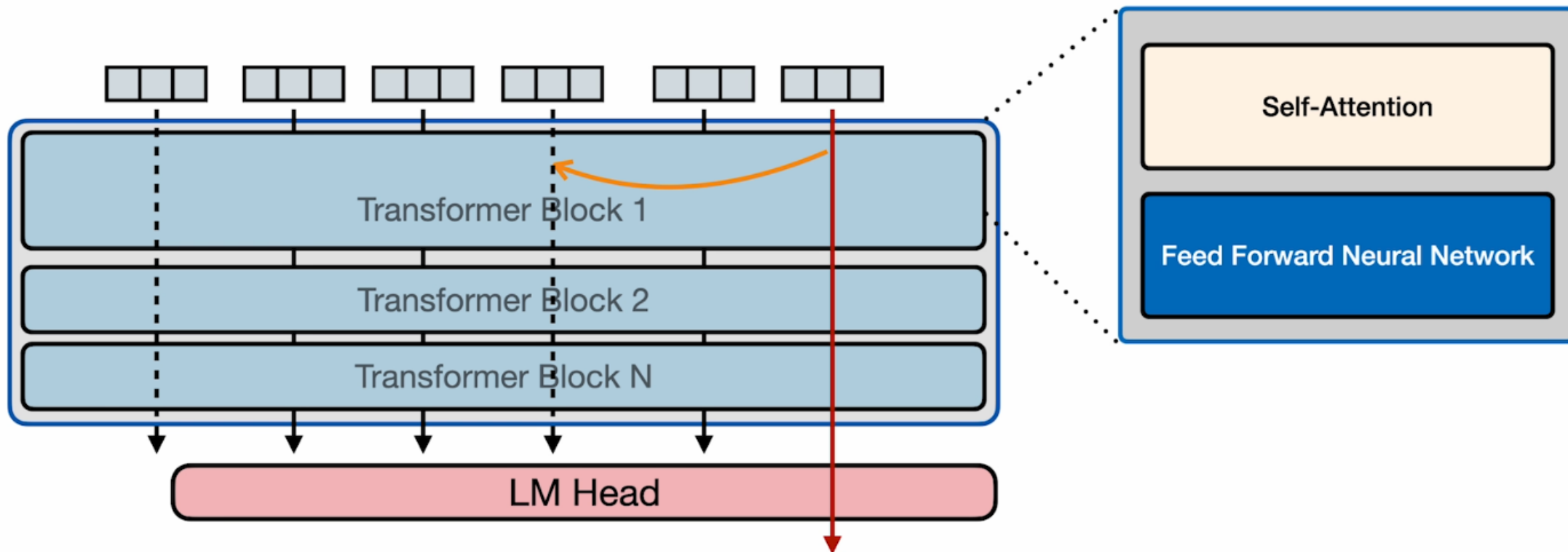


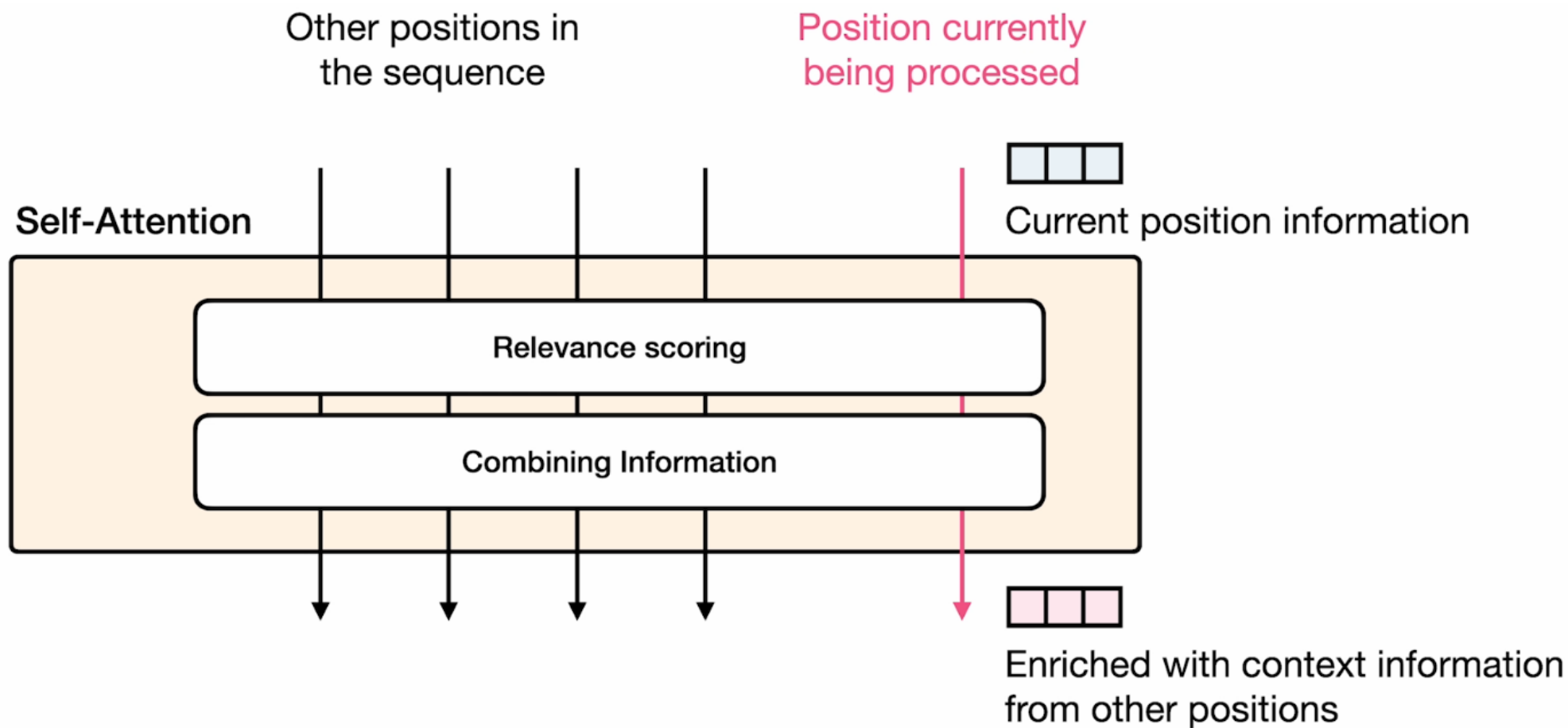
## Transformer LLM



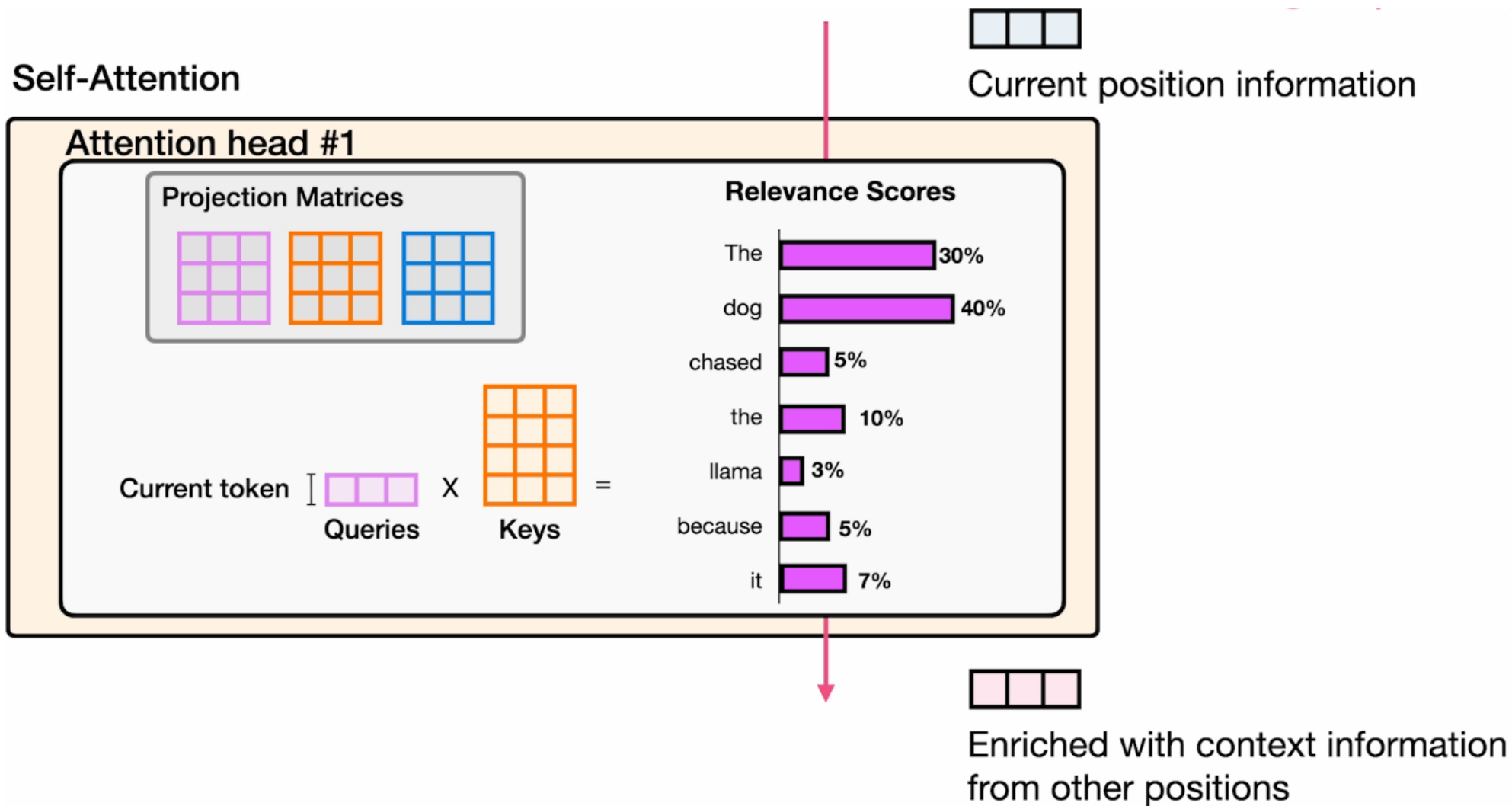


The dog chased the llama because it

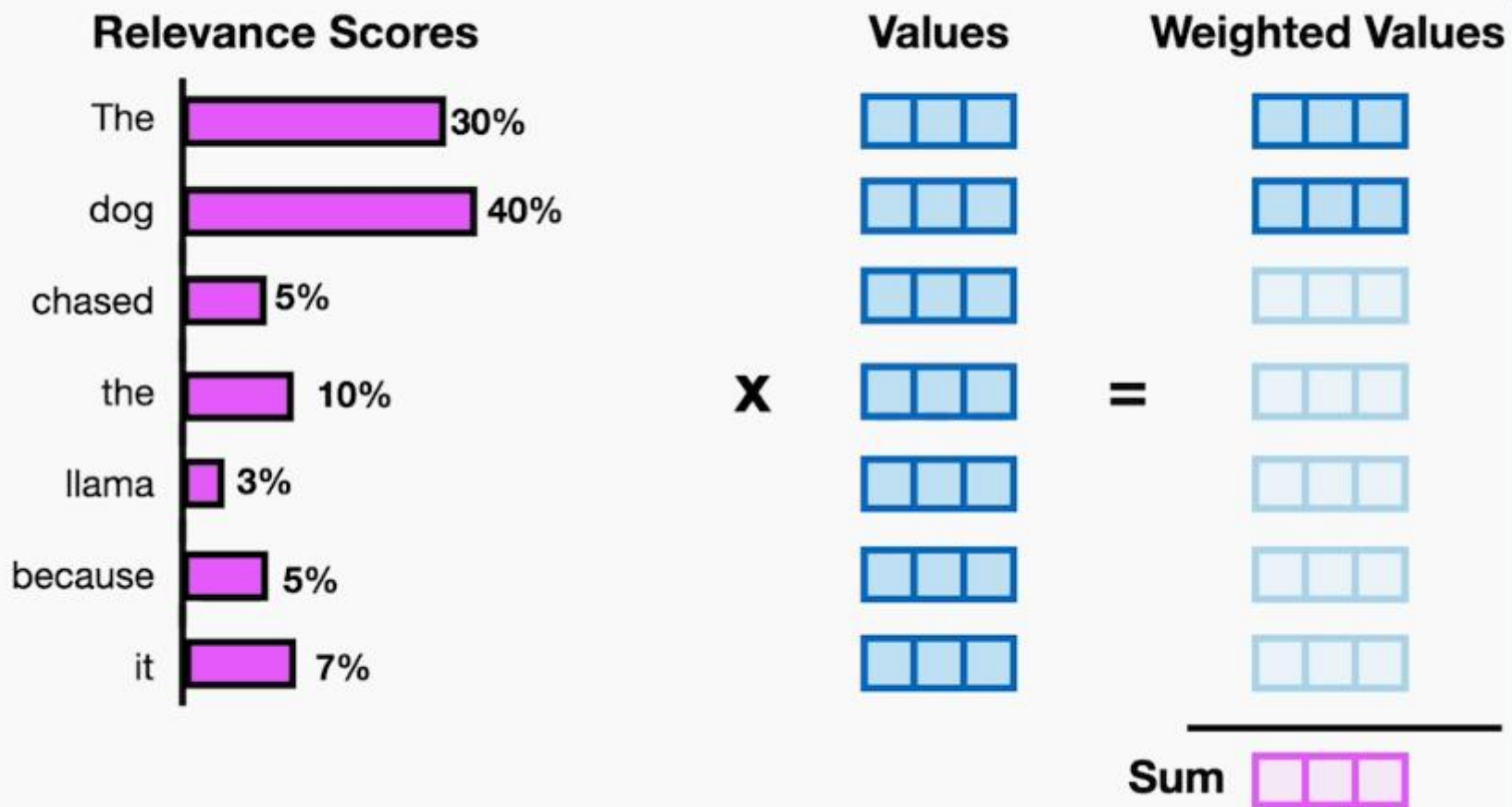




## Self-Attention



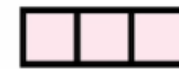
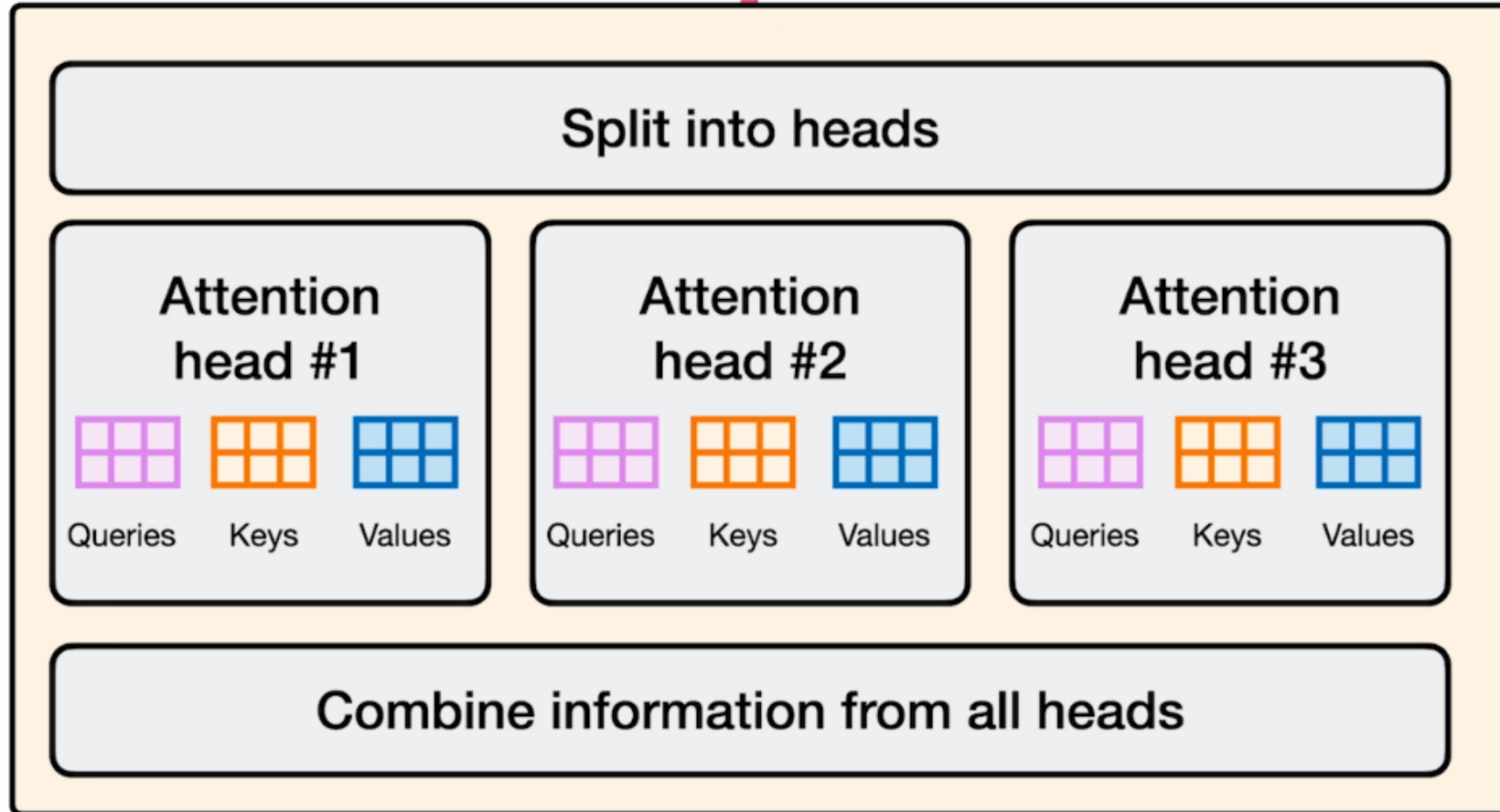
## Attention head #1



Self-Attention



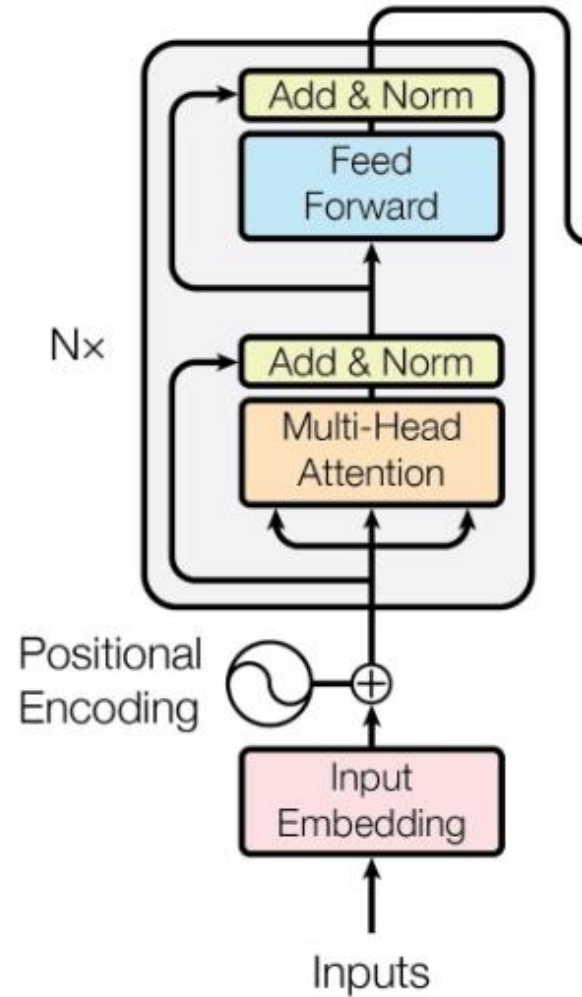
Current position information



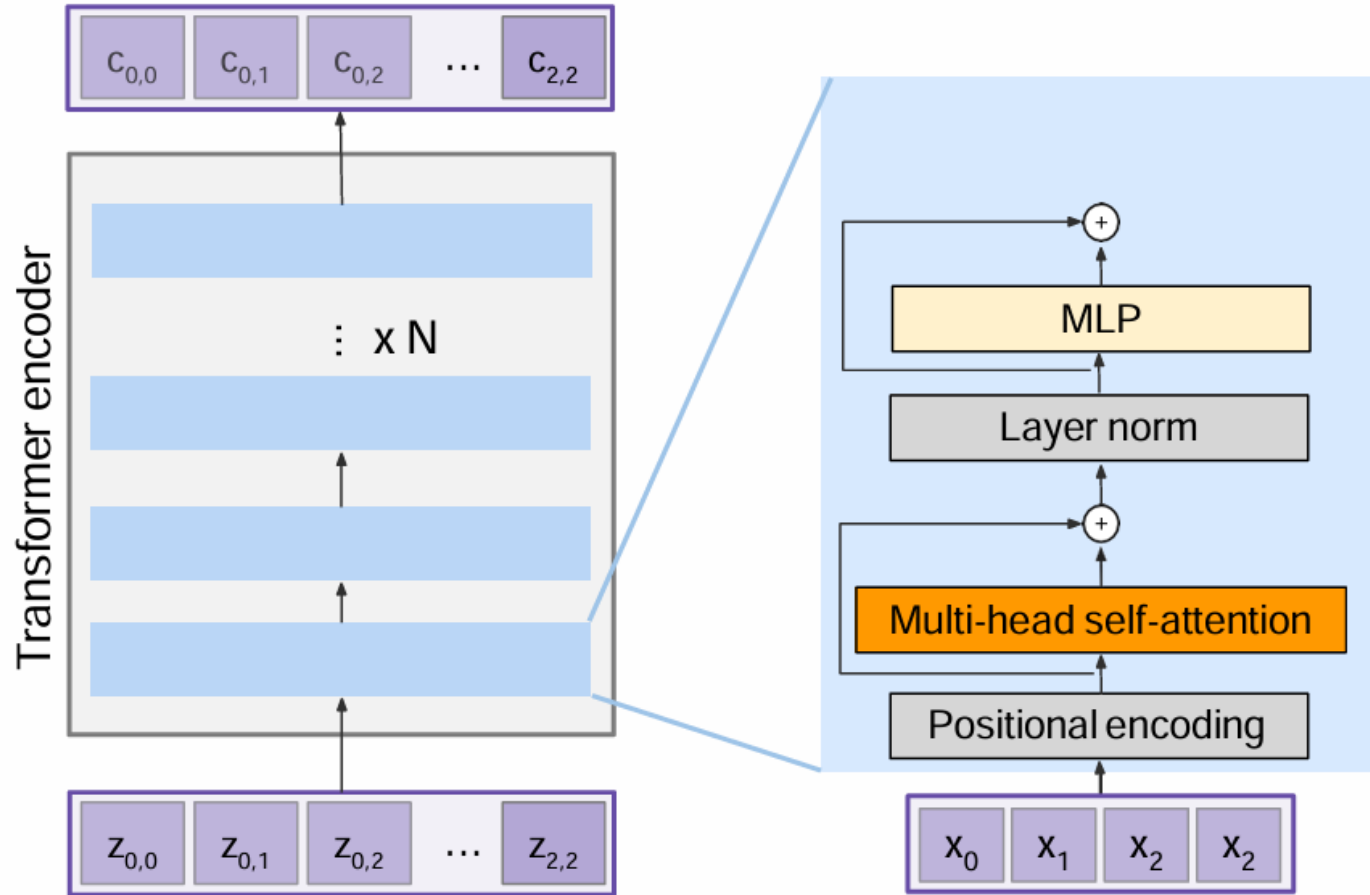
Enriched with context information  
from other positions

# Transformer Block

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# Recap



## Residual connection

MLP over each vector individually

LayerNorm over each vector individually

Residual connection

Attention attends over all the vectors

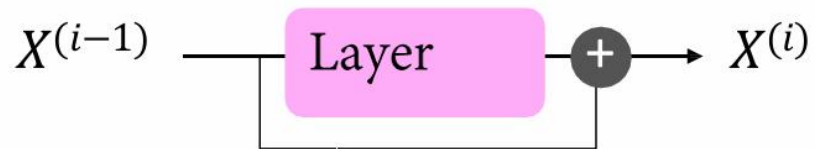
Add positional encoding

# Residual connections

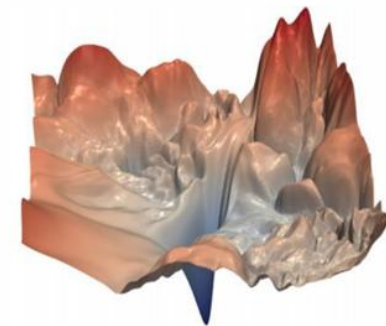
- **Residual connections** are a trick to help models train better.
  - Instead of  $X^{(i)} = \text{Layer}(X^{(i-1)})$  (where  $i$  represents the layer)



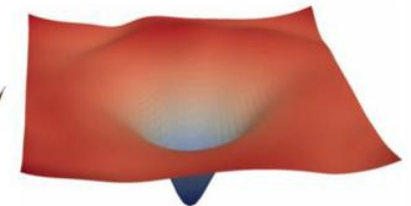
- We let  $X^{(i)} = X^{(i-1)} + \text{Layer}(X^{(i-1)})$  (so we only have to learn “the residual” from the previous layer)



- Residual connections are thought to make the loss landscape considerably smoother (thus easier training!)



[no residuals]



[residuals]

# Layer normalization

- **Layer normalization** is a trick to help models train faster.
- Idea: cut down on uninformative variation in hidden vector values by normalizing to unit mean and standard deviation **within each layer**.
  - LayerNorm's success may be due to its normalizing gradients [[Xu et al., 2019](#)]
- Let  $x \in \mathbb{R}^d$  be an individual (word) vector in the model.
- Let  $\mu = \sum_{j=1}^d x_j$
- Let  $\sigma = \sqrt{\frac{1}{d} \sum_{j=1}^d (x_j - \mu)^2}$
- Let  $\gamma \in \mathbb{R}^d$  and  $\beta \in \mathbb{R}^d$  be learned “gain” and “bias” parameters. (Can omit!)
- Then layer normalization computes:

$$\text{output} = \frac{x - \mu}{\sqrt{\sigma + \epsilon}} * \gamma + \beta$$

Normalize by scalar mean and variance

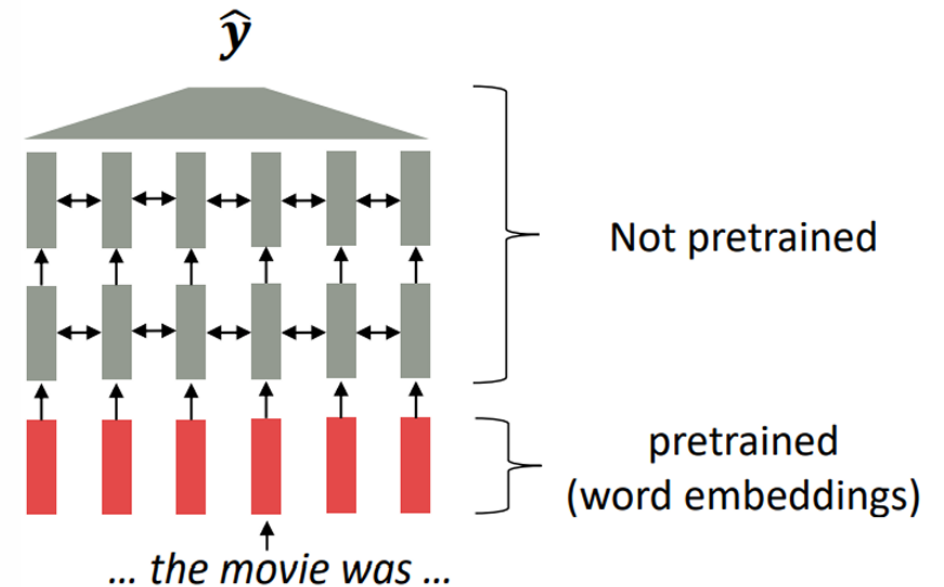
Modulate by learned elementwise gain and bias



Layer vs Batch  
Normalization

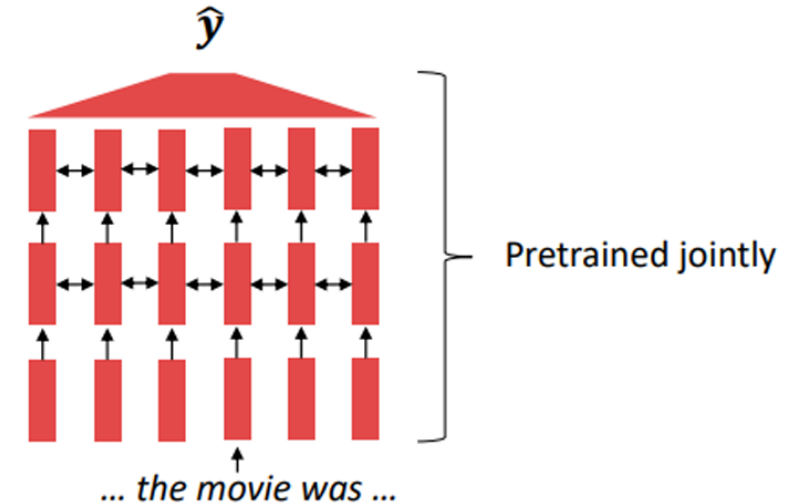
# Pretraining: Where we were

- pretrained word embeddings
- Circa 2017:
  - Start with pretrained word embeddings (no context!)
  - Learn how to incorporate context in an LSTM or Transformer while training on the task.
- Some issues to think about:
  - The training data of downstream task (like question answering) must be sufficient to teach all contextual aspects of language.
  - Most of the parameters in our network are randomly initialized!



# Transformer Block

- In modern NLP:
  - All (or almost all) parameters in networks are initialized via pretraining.
  - Pretraining methods **hide parts** of the input from the model, and train the model to **reconstruct those parts**.
- This has been exceptionally effective at building strong:
  - representations of language
  - parameter initializations for strong NLP models.
  - Probability distributions over language that we can sample from



This model has learned how to represent entire sentences through pretraining

# Masked Token Prediction

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I put \_\_\_\_ fork down on the table.

I went to the ocean to see the fish, turtles, seals, and \_\_\_\_.

The woman walked across the street, checking for traffic over \_\_\_\_ shoulder.

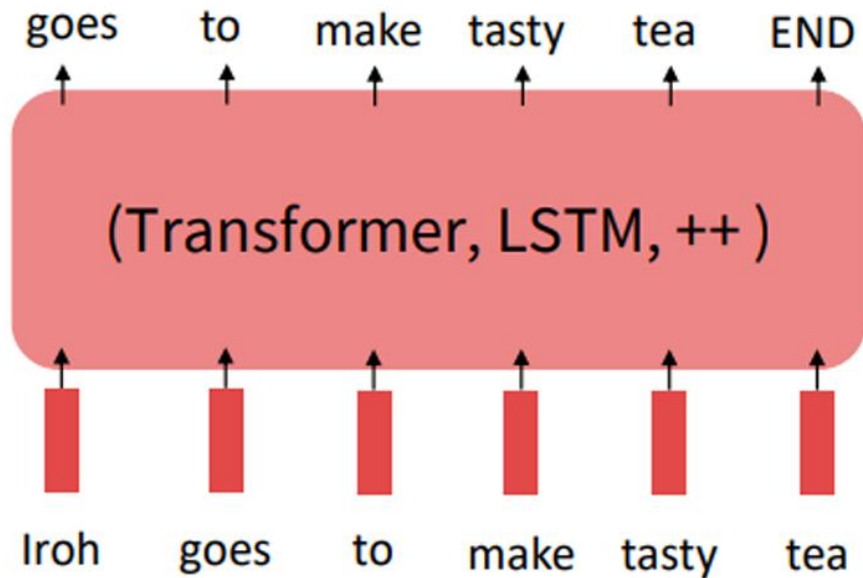
I was thinking about the sequence that goes 1, 1, 2, 3, 5, 8, 13, 21, \_\_\_\_

# The Pretraining / Finetuning Paradigm

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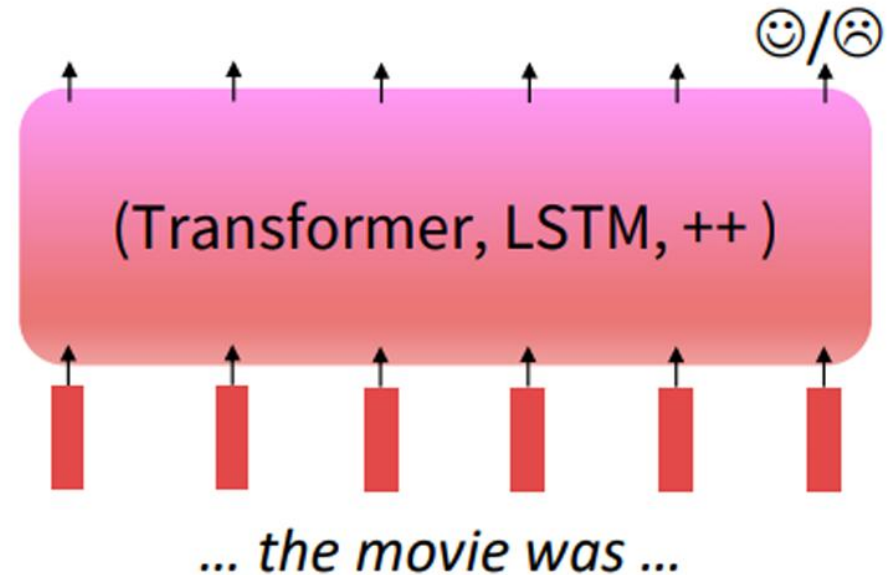
## Step 1: Pretrain (on language modeling)

Lots of text; learn general things!



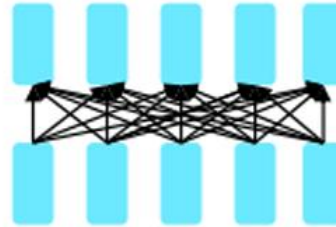
## Step 2: Finetune (on your task)

Not many labels; adapt to the task!

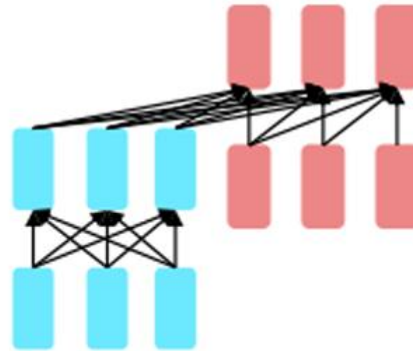


# Pretraining for three types of architectures

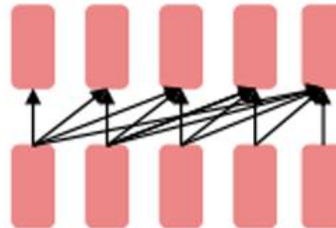
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**Encoders**



**Encoder-  
Decoders**

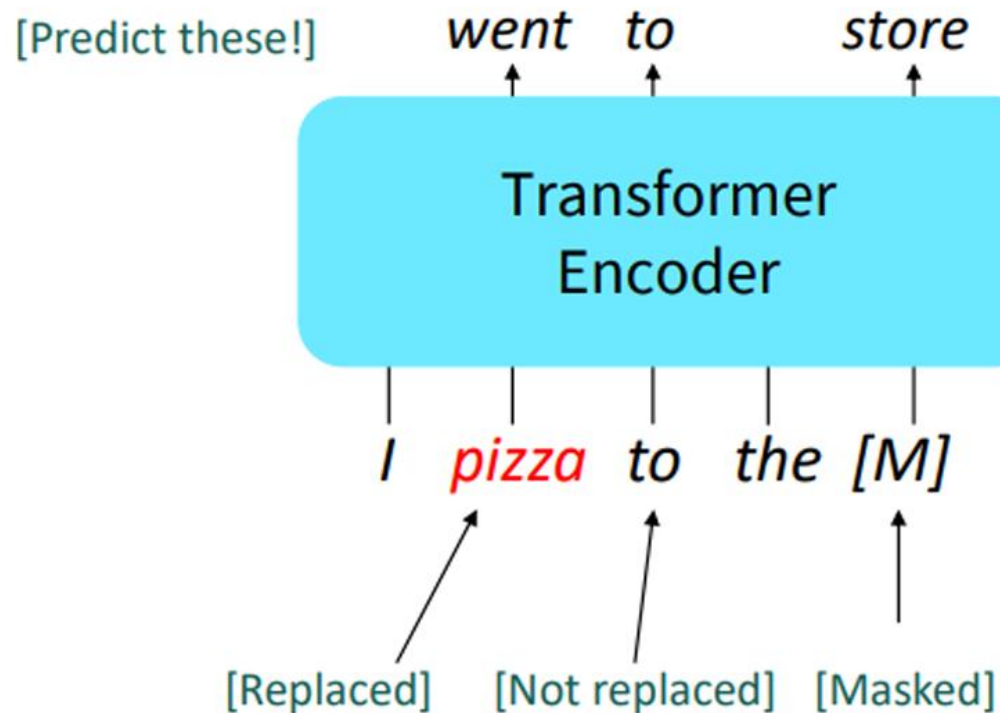


**Decoders**

# BERT: Bidirectional Encoder Representations from Transformers

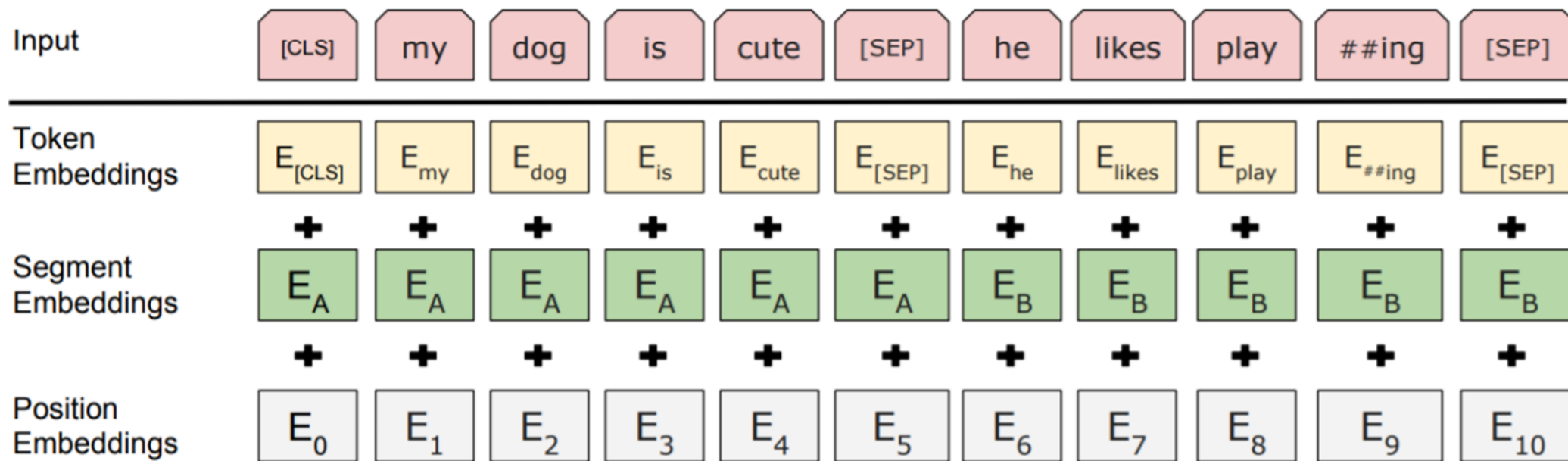
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- Devlin et al., 2018 proposed the “Masked LM” objective and released the weights of a pretrained Transformer called **BERT**.



# BERT

- The pretraining input to BERT was two separate contiguous chunks of text:



- BERT was trained to predict whether one chunk follows the other or is randomly sampled.

# BERT

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Trained on:

- BooksCorpus (800 million words)
  - English Wikipedia (2,500 million words)
- Pretraining is expensive and impractical on a single GPU.
    - BERT was pretrained with 64 TPU chips for a total of 4 days.
  - Finetuning is practical and common on a single GPU
  - “Pretrain once, finetune many times.”
  - “Small” models like BERT have become general tools in a wide range of settings

# BERT: Results

- BERT was massively popular and hugely versatile; finetuning BERT led to new state-of-the-art results on a broad range of tasks.

| System                | MNLI-(m/mm)<br>392k | QQP<br>363k | QNLI<br>108k | SST-2<br>67k | CoLA<br>8.5k | STS-B<br>5.7k | MRPC<br>3.5k | RTE<br>2.5k | Average<br>- |
|-----------------------|---------------------|-------------|--------------|--------------|--------------|---------------|--------------|-------------|--------------|
| Pre-OpenAI SOTA       | 80.6/80.1           | 66.1        | 82.3         | 93.2         | 35.0         | 81.0          | 86.0         | 61.7        | 74.0         |
| BiLSTM+ELMo+Attn      | 76.4/76.1           | 64.8        | 79.8         | 90.4         | 36.0         | 73.3          | 84.9         | 56.8        | 71.0         |
| OpenAI GPT            | 82.1/81.4           | 70.3        | 87.4         | 91.3         | 45.4         | 80.0          | 82.3         | 56.0        | 75.1         |
| BERT <sub>BASE</sub>  | 84.6/83.4           | 71.2        | 90.5         | 93.5         | 52.1         | 85.8          | 88.9         | 66.4        | 79.6         |
| BERT <sub>LARGE</sub> | <b>86.7/85.9</b>    | <b>72.1</b> | <b>92.7</b>  | <b>94.9</b>  | <b>60.5</b>  | <b>86.5</b>   | <b>89.3</b>  | <b>70.1</b> | <b>82.1</b>  |

# Extensions of BERT

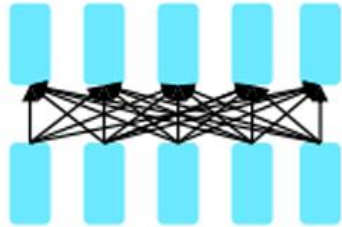
- SpanBERT
- RoBERTa: mainly just train BERT for longer and remove next sentence prediction!

| Model                    | data  | bsz | steps | SQuAD<br>(v1.1/2.0) | MNLI-m      | SST-2       |
|--------------------------|-------|-----|-------|---------------------|-------------|-------------|
| RoBERTa                  |       |     |       |                     |             |             |
| with BOOKS + WIKI        | 16GB  | 8K  | 100K  | 93.6/87.3           | 89.0        | 95.3        |
| + additional data (§3.2) | 160GB | 8K  | 100K  | 94.0/87.7           | 89.3        | 95.6        |
| + pretrain longer        | 160GB | 8K  | 300K  | 94.4/88.7           | 90.0        | 96.1        |
| + pretrain even longer   | 160GB | 8K  | 500K  | <b>94.6/89.4</b>    | <b>90.2</b> | <b>96.4</b> |
| BERT <sub>LARGE</sub>    |       |     |       |                     |             |             |
| with BOOKS + WIKI        | 13GB  | 256 | 1M    | 90.9/81.8           | 86.6        | 93.7        |

- A takeaway from the RoBERTa paper: more compute, more data can improve pretraining even when not changing the underlying Transformer encoder.

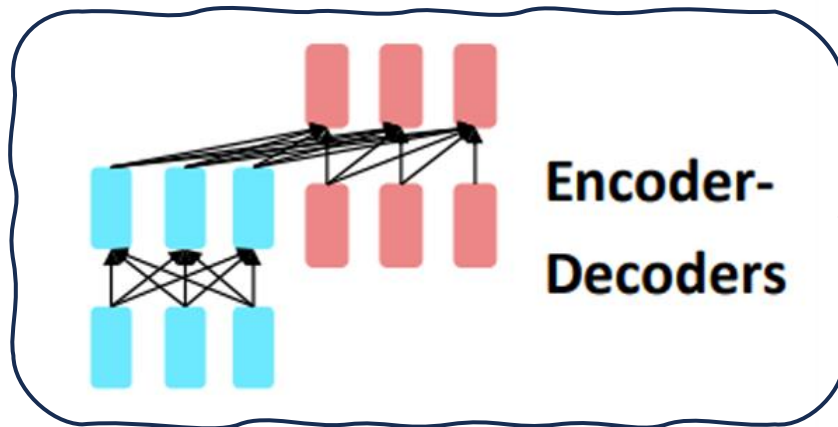
# Pretraining for three types of architectures

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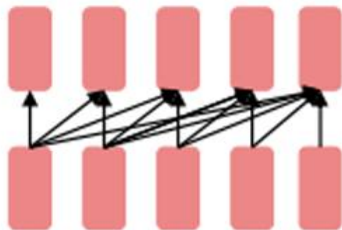


**Encoders**

- BERT



**Encoder-  
Decoders**



**Decoders**

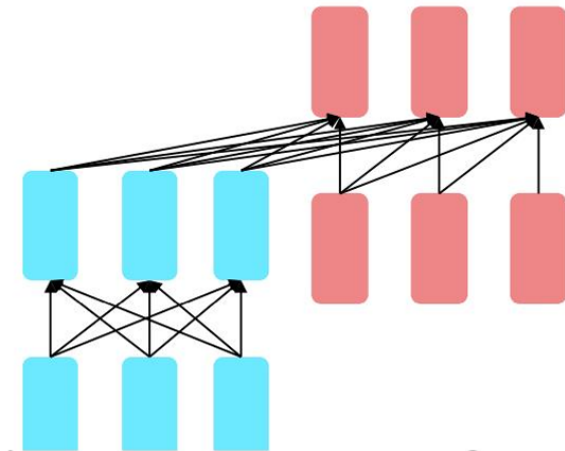
# Pretraining encoder-decoders

- **Text-to-Text Transfer Transformer (T5)** uses a text-to-text approach.
  - Every task – including translation, question answering, and classification – is cast to text-to-text.
- **T5 model:** span corruption works best.
- Replace different-length spans from the input with unique placeholders; decode out the spans that were removed!
- This is implemented in text preprocessing: it's still an objective that looks like language modeling at the decoder side.

Original text  
Thank you ~~for inviting~~ me to your party ~~last~~ week.

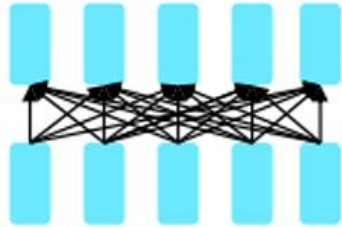
Inputs  
Thank you <X> me to your party <Y> week.

Targets  
<X> for inviting <Y> last <Z>



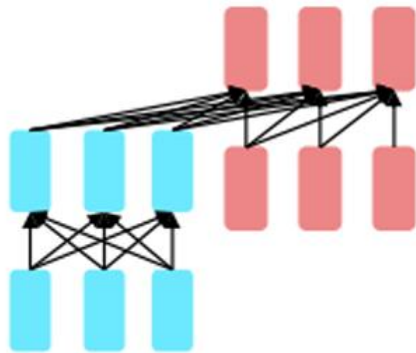
# Pretraining for three types of architectures

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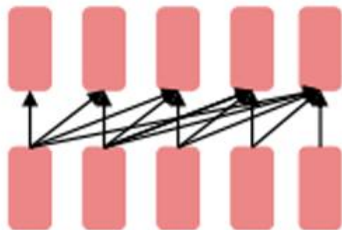
**Encoders**

- BERT



**Encoder-  
Decoders**

- T5



**Decoders**

Nice to generate from; can't condition on future words

# Generative Pretrained Transformer (GPT)

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- 2018's OpenAI GPT was a big success in pretraining a decoder!
- Transformer decoder with 12 layers, 117M parameters.
  - 768-dimensional hidden states, 3072-dimensional feed-forward hidden layers.
  - Trained on BooksCorpus: over 7000 unique books.
  - Contains long spans of contiguous text, for learning long-distance dependencies.

# GPT2

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- GPT-2, a larger version (117M->1.5B) of GPT
  - trained on much more data (4GB->40GB of WebText)
  - Scrape links posted on Reddit w/ at least 3 upvotes (rough proxy of human quality)
- It was shown to produce relatively convincing samples of natural language.

**Context (human-written):** In a shocking finding, scientist discovered a herd of unicorns living in a remote, previously unexplored valley, in the Andes Mountains. Even more surprising to the researchers was the fact that the unicorns spoke perfect English.

**GPT-2:** The scientist named the population, after their distinctive horn, Ovid's Unicorn. These four-horned, silver-white unicorns were previously unknown to science.

# GPT3

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- GPT-3 (175B parameters; Brown et al., 2020)
- Another increase in size (**1.5B** -> **175B**)
- and data (**40GB** -> over **600GB**)

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**Language Models are Few-Shot Learners**

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Tom B. Brown\*

Benjamin Mann\*

Nick Ryder\*

Melanie Subbiah\*

# Scaling Efficiency

- GPT-3 was 175B parameters and trained on 300B tokens of text.
- Roughly, the cost of training a large transformer scales as **parameters\*tokens**
- Did OpenAI strike the right parameter-token data to get the best model? No.

| Model                                              | Size (# Parameters) | Training Tokens |
|----------------------------------------------------|---------------------|-----------------|
| LaMDA ( <a href="#">Thoppilan et al., 2022</a> )   | 137 Billion         | 168 Billion     |
| GPT-3 ( <a href="#">Brown et al., 2020</a> )       | 175 Billion         | 300 Billion     |
| Jurassic ( <a href="#">Lieber et al., 2021</a> )   | 178 Billion         | 300 Billion     |
| <i>Gopher</i> ( <a href="#">Rae et al., 2021</a> ) | 280 Billion         | 300 Billion     |
| MT-NLG 530B ( <a href="#">Smith et al., 2022</a> ) | 530 Billion         | 270 Billion     |
| <i>Chinchilla</i>                                  | 70 Billion          | 1.4 Trillion    |

 This 70B parameter model is better than the much larger other models!

# GPT3

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**Fine-tuning**

**In-context learning**

# In-Context Learning (ICL)

- Given a **prompt** including:
  - An optional description of the task
  - Few-shot training examples in a prompt format demonstrating the task
  - The test input

“Apple → Red, Lime → Green, Corn →”

“Albert Einstein was German \n Mahatma Gandhi was Indian \n Marie Curie was”

## Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.

```
1  Translate English to French:  ← task description
2  cheese => .....           ← prompt
```

## Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.

```
1  Translate English to French:  ← task description
2  sea otter => loutre de mer    ← examples
3  peppermint => menthe poivrée ←
4  plush girafe => girafe peluche ←
5  cheese => .....             ← prompt
```

# ICL

---

|            |                                                         |          |
|------------|---------------------------------------------------------|----------|
| Input Text | Circulation revenue has increased by 5% in Finland.     | Positive |
|            | Panostaja did not disclose the purchase price.          | Neutral  |
|            | Paying off the national debt will be extremely painful. | Negative |
|            | The acquisition will have an immediate positive impact. | ??       |



Prediction:

Positive

# ICL

- Circulation revenue has increased by 5% in Finland. Positive
- Panostaja did not disclose the purchase price. Neutral
- Paying off the national debt will be extremely painful. Negative
- The company anticipated its operating profit to improve. ??

LM

Prediction:

Positive

- Circulation revenue has increased by 5% in Finland. Finance
- Panostaja did not disclose the purchase price. Sports
- Paying off the national debt will be extremely painful. Tech
- The company anticipated its operating profit to improve. ??

LM

Prediction:

Finance

# Prompt structure is important

- Format
- Selection
- Order

| Format ID | Prompt                                                                                                                                                                                                                                                   | Label Names        |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| 1         | Review: This movie is amazing!<br>Answer: Positive<br>Review: Horrific movie, don't see it.<br>Answer:                                                                                                                                                   | Positive, Negative |
| 2         | Review: This movie is amazing!<br>Answer: good<br>Review: Horrific movie, don't see it.<br>Answer:                                                                                                                                                       | good, bad          |
| 3         | My review for last night's film: This movie is amazing! The critics agreed that this movie was good<br>My review for last night's film: Horrific movie, don't see it. The critics agreed that this movie was                                             | good, bad          |
| 4         | Here is what our critics think for this month's films.<br>One of our critics wrote "This movie is amazing!". Her sentiment towards the film was positive.<br>One of our critics wrote "Horrific movie, don't see it". Her sentiment towards the film was | positive, negative |
| 14        | Review: This movie is amazing!<br>Positive: True<br>Review: Horrific movie, don't see it.<br>Positive:                                                                                                                                                   | True, False        |
| 15        | Review: This movie is amazing!<br>Stars: 5<br>Review: Horrific movie, don't see it.<br>Stars:                                                                                                                                                            | 5, 0               |

# Prompt structure is important

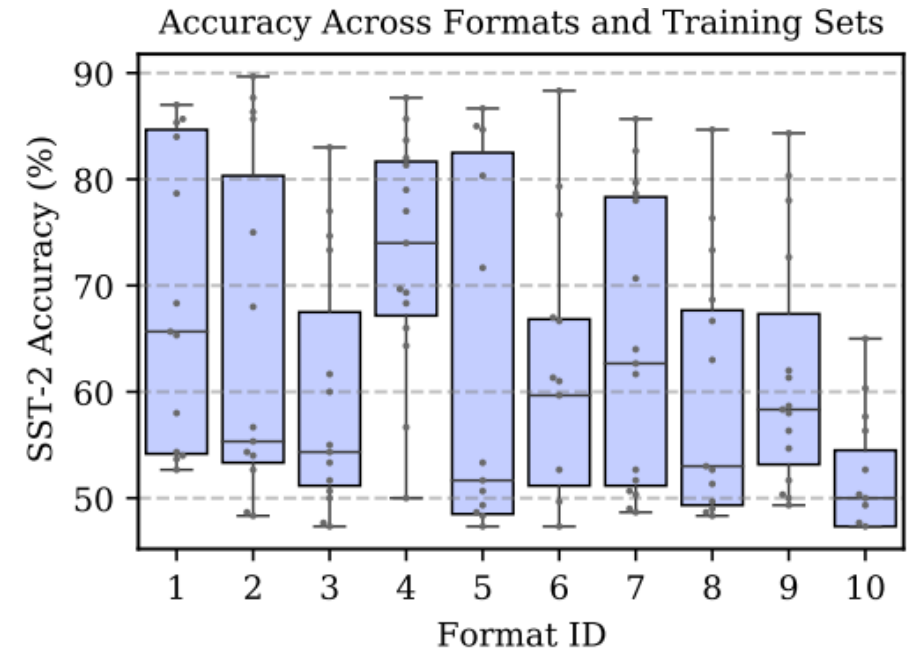
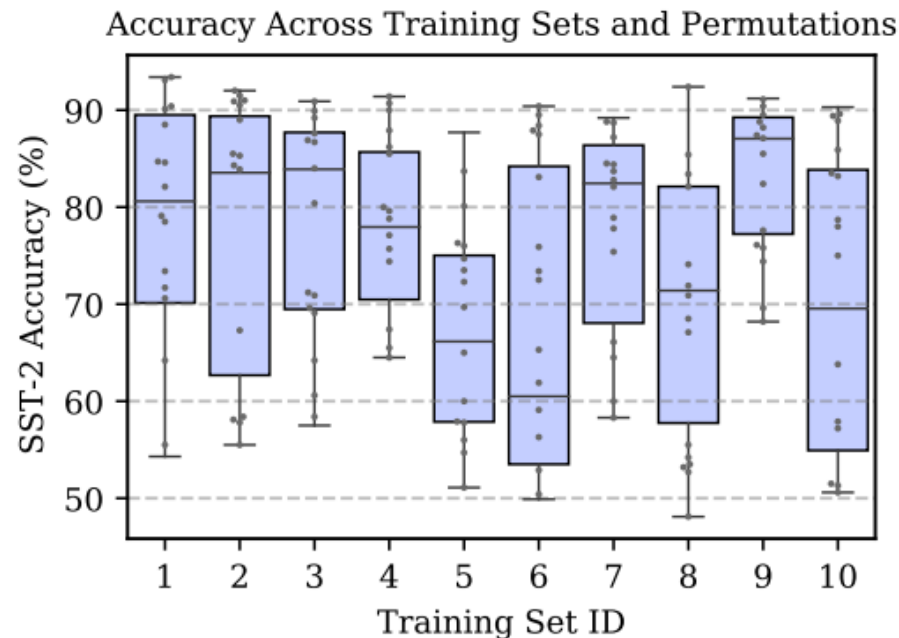
Want to read more?



Zhao, et al. **Calibrate Before Use: Improving Few-Shot Performance of Language Models** (ICML 2021)

- Format
- Selection
- Order

**ICL is highly sensitive to example permutation**



# FT vs ICL

Want to read more?



Mosbach, et al. **Few-shot Fine-tuning vs. In-context Learning: A Fair Comparison and Evaluation** (ACL 2023)

- $FT - ICL = (\text{pos}, \text{neg}, \text{zero})$

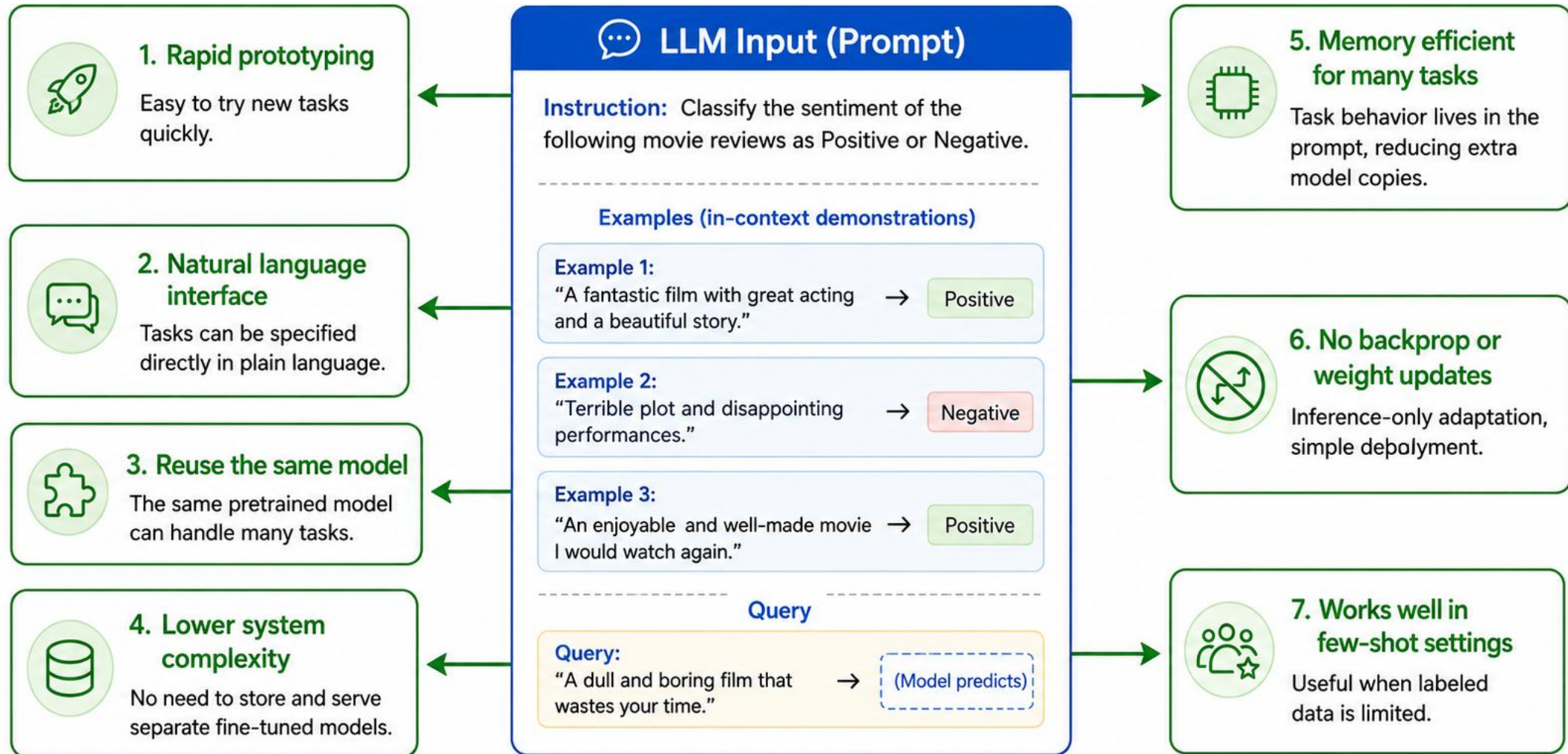
|     |      | FT    |       |       |       |      |      |       |
|-----|------|-------|-------|-------|-------|------|------|-------|
|     |      | 125M  | 350M  | 1.3B  | 2.7B  | 6.7B | 13B  | 30B   |
| ICL | 125M | −0.00 | 0.01  | 0.02  | 0.03  | 0.12 | 0.14 | 0.09  |
|     | 350M | −0.00 | 0.01  | 0.02  | 0.03  | 0.12 | 0.14 | 0.09  |
|     | 1.3B | −0.00 | 0.01  | 0.02  | 0.03  | 0.12 | 0.14 | 0.09  |
|     | 2.7B | −0.00 | 0.01  | 0.02  | 0.03  | 0.12 | 0.14 | 0.09  |
|     | 6.7B | −0.00 | 0.01  | 0.02  | 0.03  | 0.12 | 0.14 | 0.09  |
|     | 13B  | −0.04 | −0.02 | −0.01 | −0.00 | 0.09 | 0.11 | 0.05  |
|     | 30B  | −0.11 | −0.09 | −0.08 | −0.08 | 0.02 | 0.03 | −0.02 |

(a) RTE

|     |      | FT    |       |       |       |      |      |      |
|-----|------|-------|-------|-------|-------|------|------|------|
|     |      | 125M  | 350M  | 1.3B  | 2.7B  | 6.7B | 13B  | 30B  |
| ICL | 125M | −0.00 | 0.00  | 0.02  | 0.01  | 0.10 | 0.11 | 0.07 |
|     | 350M | −0.00 | 0.00  | 0.02  | 0.01  | 0.10 | 0.11 | 0.07 |
|     | 1.3B | −0.01 | −0.00 | 0.01  | 0.01  | 0.10 | 0.11 | 0.07 |
|     | 2.7B | −0.01 | −0.00 | 0.01  | 0.01  | 0.09 | 0.10 | 0.07 |
|     | 6.7B | −0.01 | −0.01 | 0.01  | 0.00  | 0.09 | 0.10 | 0.06 |
|     | 13B  | −0.03 | −0.03 | −0.02 | −0.02 | 0.07 | 0.08 | 0.04 |
|     | 30B  | −0.07 | −0.07 | −0.05 | −0.06 | 0.03 | 0.04 | 0.00 |

(b) MNLI

# ICL: Advantages



# ICL: Disadvantages

